3.5.3 Authentication factors

The primary factors can be used alone or in combination with other factors. The contextual factors are rarely used alone. We'll see in later chapters that contextual factors can also be used in a more fine-grained manner in dynamic authorization. Consequently, knowledge, possession, and inherence factors are used for direct authentication, and behavior, location, and temporal factors are typically used for risk-based authentication, strengthening security dynamically.

Each of these factors uniquely contributes to a system's ability to distinguish between legitimate requests and attacks. Let's explore how they can be used to create authentication systems that provide the needed security.

3.6 Authentication methods

When Alice lays claim to a particular account, there are four primary methods the system can use to authenticate that it's really her. All these methods use some kind of identifier that represents the claim Alice is making. These are the four methods, in increasing order of strength:

- Identifier-only
- Token-based
- Identifier with authentication factors
- Challenge-response

Let's explore each of these. In the descriptions in each section, we'll follow Alice as she authenticates using different authentication methods and see how they differ. In the figures, assume that the authentication service is part of, or is working with, some other service that Alice wants to interact with.

3.6.1 Identifier-only

In an identifier-only authentication scenario, Alice returns the same identifier with each request, thereby laying claim to an account or session. Identifier-only authentication might sound like an oxymoron. The idea of